

## **LLD (Low Level Design)**

# **Title Page**

PW SKILLS

LOW LEVEL DESIGN (LLD)

Project:

Cryptocurrency Historical Prices Prediction using Machine Learning

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## **Introduction**

The Low-Level Design (LLD) document describes the detailed implementation of the Cryptocurrency Historical Prices Prediction project. It explains each processing step, feature engineering, machine learning workflow, and prediction process.

## **Module Description**

The project consists of the following modules:

- Dataset Loading
- Data Cleaning
- Feature Engineering
- Data Preprocessing
- Data Scaling
- Train-Test Split
- Model Training
- Model Evaluation
- Prediction
- Model Saving

# Dataset Loading

The dataset was loaded using the Pandas library. The CSV file contains historical cryptocurrency prices including Open, High, Low, Close, Volume, Market Capitalization, Date, Timestamp, and Cryptocurrency Name.

# Data Cleaning

Data preprocessing included checking for missing values and infinite values. Missing values were removed where necessary, and the dataset was verified for consistency before model training.

# Feature Engineering

The following features were created to improve prediction performance:

- Price Change
- 7-Day Moving Average (MA7)
- Volatility (Target Variable)

These engineered features capture short-term market movement and price fluctuations.

# Model Development

The dataset was divided into training and testing sets using an 80:20 ratio. StandardScaler was applied to normalize numerical features. A Random Forest Regressor was trained using the processed dataset to predict cryptocurrency volatility.

# Model Evaluation

The trained model was evaluated using regression metrics including:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- $R^2$  Score

These metrics measure the prediction accuracy and overall model performance.

## Output

The project generates the following outputs:

- Predicted Volatility Values
- Model Performance Metrics
- Prediction CSV File
- Saved Machine Learning Model (.pkl)

## Conclusion

The Low-Level Design explains the detailed implementation of the machine learning pipeline. Each module works together to preprocess the dataset, train the model, evaluate performance, and generate cryptocurrency volatility predictions efficiently.